



Asset Performance Management

Calculation Logger Service

User Guide

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INTRODUCTION

The Calculation Logger runs as a windows service and records performance metrics of each calculation executed by APM.

Installation: Install the software on Calculation Server.

PRE-REQUISITES

The following are the requirements for the Calculation Logger to function:

- Microsoft .Net framework v4.8
- Communication between DataBase and calculation servers
- Access to a Microsoft SQL Server database.

INSTALLATION

4.1 Database Configuration

Perform the following steps to configure database:

1. Create a database named APMCalculationLogs.

ATTENTION

Ensure the selected data and Log directory has enough space as it will be a big database.

NOTE

All the scripts are available in DB script folder of binary file path given below:

{BuildLocation}\Utilities\Honeywell.UAS.CalculationLogging

2. Run DB script from CalculationLoggingQueriesTables.sql using SQL Server Management Studio against the created database.
3. Run DB script from CalculationLoggingQueriesProcedure.sql using SQL Server Management Studio against the created database.

4.2 Copying Binary Files

Perform the following steps to install the Calculation Logger Service:

1. In the calculation server with the Asset Performance Management media, navigate to the following folder.
<ISO>\Utilities\Honeywell.UAS.CalculationLogging\BinariesDBScript
2. Create Calculation Logger folder in one of the system drives.
3. Copy the binary files from media to the created folder.
4. Open **Honeywell.UAS.CalculationLogging.exe.config** from the created folder.
Enter the following details under application settings:
 - a. Enter the Database server name where the data base is created.
<add key="CalcHealthDatabaseConnection" value="Persist Security Info=True;Integrated Security=SSPI;Initial Catalog=SentinelCalculationLogs;server={Enter the DB server Name}"/>
 - b. Create and share the **Work Location** folder with the service user and give full permission to created work location folder.
<add key="WorkLocation" value="{Work location path}" />
5. Open command prompt in administrator mode.
6. Run the below mentioned command:

C:\Windows\Microsoft.NET\Framework64\v4.0.30319\installutil.exe "[path to Honeywell.UAS.CalculationLogging.exe]"

7. Enter domain username and password in the dialog box.

Click **Continue**.

8. The APM CalculationLogging Service is created in **Services.msc**.

NOTE

The above steps are for a single calculation server machine. In case if there are multiple devices the above process should be repeated for each individual calculation server machines.

NOTE

If you have multiple calculation server, the **value** can be any distinct word of your choice. Some of the recommended words are **First** for 1st calculation server, **Second** for 2nd calculation server, **Third** for 3rd calculation server and so on as the number of calculation server increases.

Example:

```
<add key="ServiceInstance" value="First"/>
```

4.3 Post Installation Steps

Follow the below steps in calculation server after installing calculation logger service:

1. In the calculation server with the Asset Performance Management installed media, navigate to the following folder.
<Install Location>\ProgramData\Honeywell\MES\AssetManager\AppServer\Bin
2. Open **Honeywell.AM.Service.AMHostingService.exe.config**.
3. Ensure the below appender is available and uncommented.

```
<appender name="ExternalCalcStatusAppender" type="log4net.Appender.RollingFileAppender">
  <file value="C:\ProgramData\Honeywell\AssetManager\LogFiles\CalcLogging\" />
  <appendToFile value="true" />
  <lockingModel type="log4net.Appender.FileAppender+MinimalLock" />
  <rollingStyle value="Date"/>
  <maxSizeRollBackups value="10"></maxSizeRollBackups>
  <datePattern value="'ExtStatus_'yyyyMMdd-HH:mm'.log'"/>
  <staticLogFileName value="false"/>
  <layout type="log4net.Layout.PatternLayout">
    <conversionPattern value="%d %m%n" />
  </layout>
</appender>
```


UNINSTALLATION

Perform the following steps to uninstall the Calculation Logger Service:

1. Choose **Start > Run**, type **Services.msc** and click **OK**.
The **Services** window appears.
2. In the Services panel, find **Honeywell CalculationLogger service**, right-click and select **Stop**.
3. In the calculation server, go to the installed location of calculation logger.
4. Open command prompt in administrator mode.
5. Run the below command:
C:\Windows\Microsoft.NET\Framework64\v4.0.30319\installutil.exe /u "path to Honeywell.UAS.CalculationLogging.exe"
6. The APM CalculationLogging Service is removed from **Services.msc**.

APPLICATION SETTINGS

Follow below steps to change the Application Settings parameter values:

1. Open the created Calculation Logger folder where binary files are copied.
2. Open **Honeywell.UAS.CalculationLogging.exe.config**
3. Locate **<appSettings>** section in **exe.config** file.
4. Update the value of parameters given in the below table:

Parameter	Description	Example
CalcHealthdatabase Connection	Database for calculation logging records and daily aggregation results.	<pre><add key="CalcHealthdatabase Connection" Value="Persist Security Info=True; Integrated Security=SSPI; Initial Catalog=<Database Name>; server=<DB instance name>;</pre>
CalcStatusLogLocation	The APM external calculation status log location. NOTE The file name should be ExtStatus_yyyymmdd_ HHmm.log format.	<pre><add key="CalcStatusLogLocation" value="C:\ProgramData\Honeywell\ AssetManager\LogFiles\CalcLogging"/></pre>
WorkLocation	The location where the service saves state information	<pre><add key="WorkLocation" value="C:\ProgramData\Honeywell\ UAS\WorkArea"/></pre>
DoPurgeAndArchive	To Archive and purge of old data NOTE Value is true for first calculation server and false for other servers.	<pre><add key="DoPurgeAndArchive" value="true"/></pre>
LogArchiveinDays	The log records are moved to archive table after the number of specified days.	<pre><add key="LogArchieveinDays" value="2"/></pre>
LogRetainDays	The maximum number of days the log records need to be retained. After the specified number of days	<pre><add key="LogRetainDays" value="10"/></pre>

Parameter	Description	Example
	the records are removed.	
ReadFrequencyMinutes	The time interval for the service to parse the log files.	<add key="ReadFrequencyMinutes" value="1"/>
CalcStatFrequencyMinutes	The time interval for the service calculations daily aggregation results.	<add key="CalcStatFrequencyMinutes" value="120"/>
ScheduleCalcDelayMinutes	The scheduled calculation is counted as delayed, if the difference between start time and schedule time exceeds the specified value.	<add key="ScheduleCalcDelayMinutes" value="30"/>
DataDrivenCalcDelayMinutes	The data-driven calculation is counted as delayed, if the difference between the start time and latest time of input exceeds the specified value.	<add key="DataDrivenCalcDelayMinutes" value="30"/>
StatRetainDays	The records older than the specified number of days are deleted.	<add key="StatRetainDays" value="35"/>
CalcStatWarmupMinutes	CalcStat processing will wait for WarmupMinutes after service start.	<add key="CalcStatWarmupMinutes" value="120"/>
CalcLogBatchSize	The batch size of CalcStat	<add key="CalcStatBatchSize" value="1000"/>
CalcLogBatchSize	The batch size of Calculation log.	<add key="CalcLogBatchSize" value="10000"/>

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